

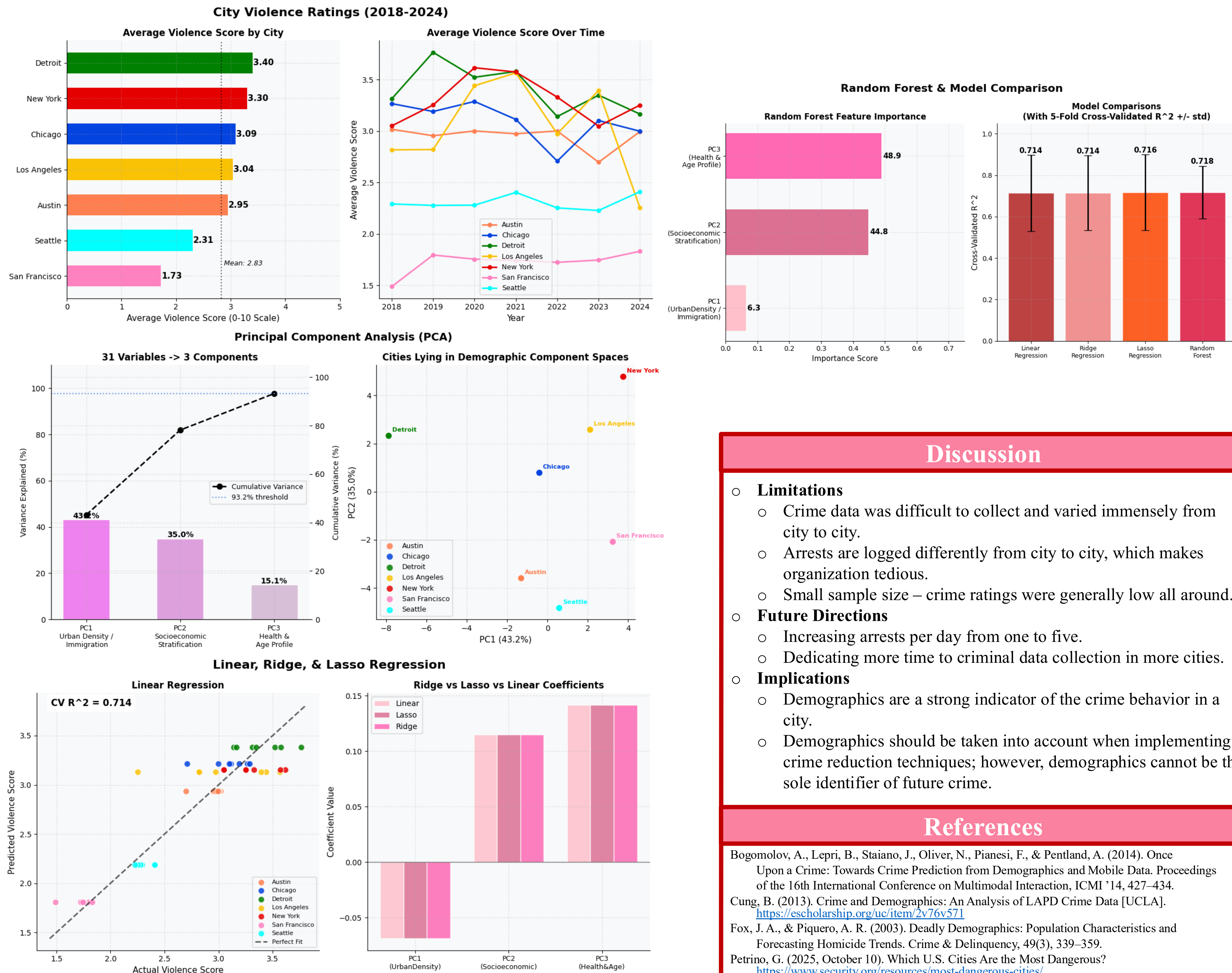
Introduction

- In recent years, there has been a decrease in crime in the United States, but the country's violent crimes are still significantly higher than those of other countries.
- Demographic change was found to be an important contributor to a downward slope in crime (Fox & Piquero, 2003).
- Human behavior and demographics have definitively been used to predict crime (Bogomolov et al., 2014).
- Los Angeles uses techniques and technology that help predict crime and allocate resources to crime-ridden areas (Cung, 2013).
- **Goal**
 - This project aims to provide a tangible prediction model to help determine a city's violent crime rating based on its demographic makeup.

Data Collection & Manipulation

- Analyzed data from the United States Census and open government crime data from seven major US cities (NYC, LA, Chicago, Austin, Seattle, Detroit, and San Francisco).
- Randomly selected one arrest per day from 2018 to 2024 for each city.
- Originally included 31 demographic variables.
- Assigned arrests to an arrest group and gave it a violence rating out of 10.
- Aggregated data on the average violence rating per year, per city.
- Took the logarithmic values of highly skewed variables.
- Standardized all values before analyses.
- **Principal Component Analysis**
 - PCA was used to determine the top 3 components for city demographics that explained the most variation in the model (93.2% of variation).
 - **PC1:** Urban Density/Immigration
 - # of Asians, Foreign-born rate, Homeownership rate
 - **PC2:** Socioeconomic Stratification
 - # of Blacks or African Americans, Graduate or Professional degree rate, Education rate
 - **PC3:** Health & Age
 - Healthcare rate, Older population rate, mean Age
- Linear regression, ridge regression, lasso regression, and a Random Forest were run to determine model efficiency and component importance.
- The final prediction model was built using Random Forest.

Visualizations



Discussion

- **Limitations**
 - Crime data was difficult to collect and varied immensely from city to city.
 - Arrests are logged differently from city to city, which makes organization tedious.
 - Small sample size – crime ratings were generally low all around.
- **Future Directions**
 - Increasing arrests per day from one to five.
 - Dedicating more time to criminal data collection in more cities.
- **Implications**
 - Demographics are a strong indicator of the crime behavior in a city.
 - Demographics should be taken into account when implementing crime reduction techniques; however, demographics cannot be the sole identifier of future crime.

References

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